1. Acids, Bases and Salts

1.1. Properties of Acids

Acid: a substance that produces hydrogen ions (H^+) when dissolved in water.

Definition of Acids: **Proton donors**.

Indicators

- Have a pH between 1 (strong) and 6 (weak)
- Turns blue **litmus paper** red
- Turns **methyl orange** indicator red
- Colourless in **Thymolphthalein**

Weak and Strong Acids

1. **Strong acids: completely dissociated in aqueous solution** producing lots of H^+ ions

e.g: HCl (aq) → H^+ (aq) + Cl^- (aq)

2. **Weak acids: partially dissociated in an aqueous solution** producing fewer H^+ ions

e.g: CH_3COOH (aq) ⇌ H^+ (aq) + CH_3COO^- (aq)

Chemical properties

- Acid + metal → salt + hydrogen gas
- Acid + base → salt + water
- Acid + metal carbonate → salt + carbon dioxide + water

1.2. Properties of Bases

Bases: substances which neutralise acids to form salt and water only.

Definition of Bases: They are **proton acceptors** (form OH^- ions)

Indicators

- Have a pH between 8 (weak) and 14 (strong)
- Turns red **litmus** paper blue
- Turns **methyl orange** indicator yellow
- Turns Blue in **thymolphthalein**

Weak and Strong Alkalis (Soluble Bases)

1. **Strong alkalis:** completely dissociates in aqueous solution, producing lots of OH^- ions

e.g.: NaOH (aq) → Na^+ (aq) + OH^- (aq)

2. **Weak alkalis** partially ionize in water producing OH^- ions

e.g: NH_4OH (aq) ⇌ NH_4^+ (aq) + OH^- (aq)

Chemical properties

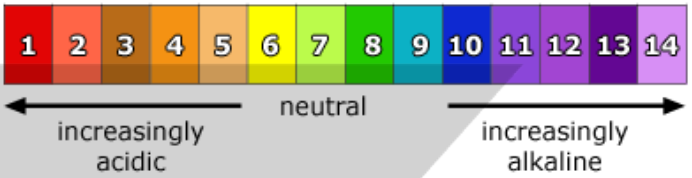
- Base + acid → salt + water (+ CO_2 when base is a metal carbonate)
- Base + ammonium salt → salt + ammonia gas + water

1.3. Neutral

- Neutral substances have a pH value of 7.
- **pH scale**

  pH is the concentration of H^+ ions per dm^3 of solution

  Universal indicator solution is used to determine the pH of a substance by matching the color change to the pH color chart.



1.4. Types of Oxides

- Metal oxides are **basic,** e.g. Copper oxide and Calcium oxide
- Non-metal oxides are **acidic,** e.g. sulfur dioxide and carbon dioxide
- Aluminium and zinc form **amphoteric oxides,** e.g. zinc oxide
- Reaction with Acid:
  - $ZnO + 2HCl \rightarrow ZnCl_2 + H_2O$
  - $Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$
- Reaction with Base:
  - $ZnO + 2NaOH \rightarrow Na_2ZnO_2 + H_2O$
  - $Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$
- Oxides that react with neither acids nor bases are **neutral,** e.g. nitrous monoxide and carbon monoxide

Colours of Transition Metal Compounds

Metal Compounds	Colour
Copper (II) Sulfate	Blue
Copper (II) Oxide	Black
Copper (II) Carbonate	Green
Manganese (IV) Oxide	Black
Iron (II) Salts	Pale Green
Iron (III) Salts	Brown or Orange

1.5. Preparation of Salts

Soluble Salts

Method A: Soluble Salts from Excess Insoluble Bases (metal, metal oxide, carbonates)

1. Warm acid (increases the speed of reaction)

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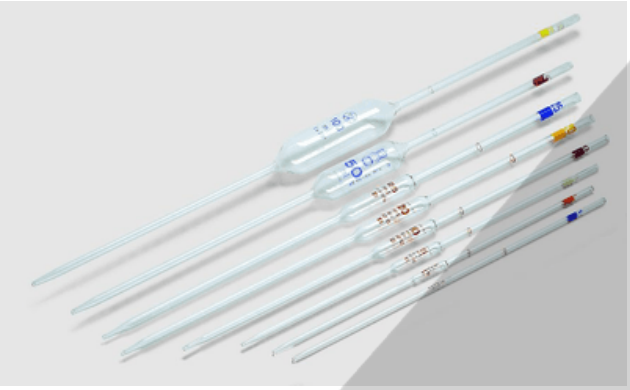
2. Add an excess reactant + stir.
3. Filter mixture
4. Transfer to evaporating basin
5. Heat using a Bunsen burner
6. Leave to cool until crystallisation point
7. Dry crystals on filter paper

Method B: Titration

1. Place a known volume of alkali into a conical flask using a volumetric pipette.
2. Add indicator (e.g. thymolphthalein)
3. Titration: add acid using a burette until the endpoint has reached
4. Record the volume of acid added
5. Repeat without indicator
6. Transfer to evaporating dish
7. Heat with Bunsen burner
8. Leave to cool until crystallisation point
9. Wash crystals with distilled water
10. Dry crystals on filter paper



Insoluble Salts

Precipitation: A reaction in which an insoluble solid product is formed when two solutions are mixed together.

1. Mix two soluble salts
2. Filter to remove the precipitate
3. Wash the precipitate with distilled water
4. Leave to dry

1.6. Water of Crystallisation and General Solubility Rules

Water of Crystallisation

- A salt is a compound formed when a metal replaces all the hydrogen atoms of an acid.
- Naming salts involves two parts: the name of the metal and the acid-ending

  eg. calcium + hydrochloric acid = calcium chloride
- The Water of Crystallisation is the **fixed number of water molecules present within a hydrated crystal.**

Type of Salt	Acid used
Sulfate	Sulfuric acid
Nitrate	Nitric acid
Chloride	Hydrochloric acid
Ethanoate	Ethanoic acid

General Solubility Rules

Soluble Salts	Insoluble Salts
All sodium, potassium and Ammonium salts	None
All nitrates	None
Chlorides	Except for silver and lead
Sulfates	Except for barium, lead and calcium
Potassium, Sodium and Ammonium Carbonates	All other carbonates
Sodium, Potassium and Ammonium Hydroxides (partially calcium hydroxide)	Nearly all hydroxides

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